PLACE VALUE STUDIO & REGROUPING TRAYS

Dual Whole Number & Decimal Activity Trays with Base-Ten Regrouping Tokens

SERIES 1 • CARD 12 / 12 • DOMAIN: POSITIONAL BASE-10 & REGROUPING

CARD 12 / 12 • PLACE-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

POSITIONAL BASE-10 NOTATION & REGROUPING

A digit's value depends on its place column: $d_k \times 10^k$.
Addition: Trade 10 units for 1 Carry block (+1) one column left.
Subtraction: Borrow 1 from left column for a (10) Trade block.
Dual 5x2 trays connect whole numbers directly to decimals!

CALIBRATED 19.00 MM POSITIONAL GRID

- Digit Tile: 19.00 × 19.00 × 7.00 mm (1U) with 1.85mm detent
- Whole Tray: 10,000, 1000, 100, 10, 1 (Inner 5×2 Tray #1)
- Decimal Tray: 0.1, 0.01, 0.001, 10^{-4}, 10^{-5} (Inner 5×2 Tray #2)

PLACE VALUE REGROUPING: 347 + 285 = 632

THOU	HUND	TENS	ONES
	3	+1 4	7
	2	8	5

REGROUPING TOKENS

Trade 10 Ones for 1 Ten
7 + 5 = 12 → 2 ones, +1 carry

Dual 6x2 Trays:

Whole: Thousands to Ones
Decimal: Ones to Thousandths

2. KIT PARTS INVENTORY (56 PIECES)

COMPONENT	QTY	SPEC	COLOR
White Digits (0–9)	22x	4×0, 2×1–9	PLA White
Blue Digits (0–9)	22x	4×0, 2×1–9	Blue-Gray
Carry & Borrow Tokens	8x	4×, 4×	Glow Yellow
Whole Numbers Tray	1x	Inner 5×2	Blue-Gray
Decimal Numbers Tray	1x	Inner 5×2	Blue-Gray
3x1 Storage Bin	1x	49.0 mm	PLA White
[123] Docking Lid	1x	Docking Lid	PLA White

TOTAL: 44 Digit Tiles + 8 (+1/10) Tokens + 2 Inner 5x2 Trays + Box + Lid = 56 Parts

SYMMETRIC 2-COLOR OPERAND & REGROUPING PALETTE

- PLA White (Set A, 22x): 4x '0' + 2x each of '1'–'9' + Box + Lid.
- Slate Blue-Gray (Set B, 22x): 4x '0' + 2x each of '1'–'9'.
- Slate Blue-Gray (2x Inner Trays): 5×2 Whole & Decimal Trays.
- Glow Yellow / Red (8x): 4x (+1) Carry + 4x (10) Borrow tokens.

3. PACK-UP & SUPPORT

✓ **10-SECOND PACK-UP**

Stack 4 blocks in Cols 0–4 (0–9), stack 6 blocks in Col 5 (+1, 10, & extra 0s = 52 blocks total), nest the 2 inner 5x2 trays over Cols 0–4 (leaving Col 5 for easy finger grip)

✓ **NO LOST PIECE GUARANTEE**

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ **HANDS-ON TUTORIAL**

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Every real number has a unique base-10 positional representation.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: MULTI-DIGIT ADDITION & DECIMAL REGROUPING

Step-by-Step Hands-On Tutorial: 347 + 285 = 632 with Regrouping Tokens

SERIES 1 • CARD 12 OF 12 • DOMAIN: POSITIONAL BASE-10 & REGROUPING

CARD 12 / 12 • PLACE-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/place-value/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 2.NBT.A.1 (Place Value)
- CCSS 4.NBT.B.4 (Addition/Sub...)
- CCSS 5.NBT.A.3 (Decimals to ...)

Grades 1–5 Elementary Mathematics

GUIDING INQUIRY

What happens when the counters in any column reach ten?

① STEP 1: POPULATE OPERANDS 347 AND 285

Place 347 in top row (White) and 285 in bottom row (Tan).
Align 3 hundreds, 4 tens, 7 ones over 2 hundreds, 8 tens, 5 ones.
Visual clarity: operand contrast eliminates confusion.

② STEP 2: REGROUP ONES (7 + 5 = 12)

7 + 5 = 12. Trade 10 ones for a Red (+1) token.
Place the Red token on the Tens column digit block.
Keep 2 in the Ones column.

③-④ STEPS 3 & 4: REGROUP TENS & COMPLETE

3. REGROUP TENS: 4 + 8 + 1 = 13. Trade 10 tens for a Red (+1) in Hundreds.
4. READ COMPLETED SUM: Keep 3 in Tens; Hundreds: 3 + 2 + 1 = 6. Total = 632!

⑤ CORE PRINCIPLE: CARRYING IS PHYSICAL TRADING

In standard algorithms, carrying is a mysterious tiny 1 written above a column.
Here, carrying is a physical trade: 10 units become 1 ten block.
Students see that value is conserved across columns!

★ EXTENSION INVESTIGATION: DECIMAL CHALLENGE: 1.75 + 0.85

Set up Decimal tray with 1.75 and 0.85. Regroup hundredths and tenths. What is the sum?
2.60! (Scan QR for 3D lesson!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

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Interactive Mathematics Manipulative Ecosystem • Mountain View, CA • BPA-Free

PARTS & REFILL PORTAL
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STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

PLACE-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 56 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

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